Sanika Mhatre

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Education

Year	Degree	University	%/CGPA
2024 - 2026	M.Tech Data Science	Manipal Institute of Technology, MAHE, Manipal	**8.99/10**
2020 - 2024	B.Tech Electronics and Computer Science *Specialization in Data Analytics and Cloud Computing*	Pillai College of Engineering, New Panvel	**8.71/10**

Technical Skills

- **Programming Languages:** Python, R Programming, SQL
- **Core Concepts:** Machine Learning, Deep Learning, Large Language Models, Retrieval-Augmented Generation, MLOps/LLMOps, AI Agents, Prompt Engineering, GenAI, Data Science, Data Analysis, Artificial Intelligence
- **Databases:** MySQL, PostgreSQL
- **Cloud Platforms:** AWS, GCP, Azure
- **Tools & Frameworks:** Flask, Docker, Git / GitHub, Power BI, Tableau

Experience

Philips June 2025 - Sept 2026

AI and Data Science Intern

- Built reusable MLOps/LLMOps pipelines for distributed GPU training, evaluation, versioning, and lifecycle automation, cutting manual orchestration by **80%** and speeding up reproducible model onboarding.
- Developed end-to-end classical ML pipelines and production-grade 2D/3D deep learning models on 10,000+ DICOM images, using XAI techniques to improve interpretability for clinical validation and deployment.
- Architected GenAI, agentic AI, and RAG pipelines for grounded natural-language querying over enterprise data, cutting lookup time by **70%** and speeding up decisions.
- Built a real-time analytics dashboard with 4 KPI categories, 12 interactive visualizations, automated reporting, and record-level drill-down, cutting manual reporting by **60%** and saving 3–5 hours weekly.

TCIAGLOBAL January 2024 - April 2024

Trainee Cloud Engineer

- Developed a cloud-based geo-routing solution with traffic management and load balancing, dynamically directing users to location-appropriate endpoints to optimize latency, availability, and user experience.
- Supported end-to-end Azure cloud operations across diverse infrastructure services, handling provisioning, monitoring, troubleshooting, and incident resolution for enterprise environments, ensuring service reliability and continuity.

Projects

Kidney Tumor Segmentation

- Developed a double U-Net++ architecture for kidney tumor segmentation, replacing the default encoder with a pretrained EfficientNet-B1 backbone to leverage transfer learning for small tumor feature extraction.
- Improved small tumor localization by combining U-Net++ nested skip connections with EfficientNet-B1's multi-scale feature maps, enhancing boundary precision on low-contrast lesions.
- Achieved a Dice score of 92%, validating the effectiveness of the encoder swap on medical image segmentation benchmarks.

Clinical RAG Assistant

- Developed a Clinical Notes Q&A system to help healthcare professionals instantly query medical knowledge across 40+ specialties, replacing manual document search with an AI assistant grounded in real medical transcriptions.
- Built using hybrid RAG combining semantic and keyword search with cross-encoder reranking, with production features including PII redaction, conversational memory, and specialty-aware retrieval.
- Evaluated system quality using RAGAs with an independent judge LLM to eliminate self-evaluation bias, tracked via LangSmith observability and a custom HITL feedback dashboard.

Publications

A Comparative Analysis of Optimized Machine Learning Pipeline with Explainable AI for Sleep Disorder Classification IEEE CONECCT 2026

- Engineered an end-to-end explainable ML pipeline for automated sleep disorder screening, achieving state-of-the-art 98.67% accuracy while solving the clinical black-box problem, making AI-driven diagnosis both accurate and trustworthy for real-world healthcare deployment.